Rishabh Shah

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Education

Rutgers University, MS in Computer Science. — Sep. 2025 – May. 2027
- **Coursework:** Data Structures & Algorithms, Operating Systems, Introduction to AI.

University of Mumbai, B.Tech in AI & Data Science, **Honors** in Computational Biology. — Dec. 2021 – May. 2025
- **GPA:** 3.25 / 4.0
- **Coursework:** Data Structures & Algorithms, Operating Systems, Artificial Intelligence, Cloud Computing.

Technical Skills

- **Programming & Scripting:** Python, Java, C++, C, SQL
- **Full-Stack Development:** Next.js, Flask, Django, FastAPI, REST APIs, Microservices, Authentication (JWT)
- **Databases & Data Handling:** MySQL, PostgreSQL, MongoDB, Schema Design, Query Optimization, Data Modeling
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, CI/CD (GitHub Actions), Git, Linux/Unix, System Monitoring
- **Additional Technologies:** TensorFlow, PyTorch, YOLOv12, OpenCV, OCR/ANPR

Experience

Aaditya Technologies, Mumbai, India — Jan. 2025 – Jul. 2025
Software Engineering Intern
- Engineered and implemented backend services using Python, Next.js, and MySQL, developing modular APIs, refining database schema, and optimizing queries to improve data handling efficiency and reduce average response time by 22%.
- Rolled out secure REST APIs with JWT-based authentication, enabling seamless communication between modules, integrating error-handling and logging systems, and enhancing data reliability by 32% while reducing integration issues.
- Containerized and deployed scalable microservices using Docker and AWS, automating CI/CD workflows with GitHub Actions, improving deployment speed by 45%, ensuring consistent environments, and minimizing downtime during updates.

Creative Line, Mumbai, India — Jun. 2024 – Sep. 2024
Web Developer Intern
- Optimized a core Next.js app, reducing page load time by 18%, enhancing responsiveness via code-splitting, asset compression.
- Automated key data flows by building and testing Next.js API routes, reducing manual operations by 38%, and accelerated data retrieval by 23% through MySQL query optimization.
- Conducted testing, debugging, and documenting code, improving overall application reliability and helping reduce development issues by 15% and creating reusable documentation to enhance team workflow and efficiency.

Technical Projects

Traffic Violation Detection & Automated Ticketing System — github.com/RishiiShah
- Developed a YOLO-based computer vision pipeline detecting five traffic violations (speeding, red-light, no-helmet, expired PUC/insurance) with 95% accuracy and 18% fewer false positives than baseline.
- Engineered backend automation for challan (ticket) generation and delivery using AWS S3 (PDF storage) and Twilio API (SMS notifications), reducing manual workload by 90%.
- Integrated OCR (Llama inference) for license-plate recognition and streamlined preprocessing (tripwire logic, frame handling) for 30% faster inference and scalable edge deployment.

JARVIS: Voice Assistant with Smart Home Automation — github.com/RishiiShah/Jarvis
- Developed a full-stack IoT automation system with voice and web control, using Django, Azure IoT Hub, and Raspberry Pi.
- Integrated Llama-3 (70B) via Groq API, achieving below 300 ms latency and 40% higher throughput with asynchronous I/O.
- Designed REST APIs for secure device communication, ensuring 99.9% state consistency and real-time cloud–local sync.

Music Genre Detection (ML Pipeline) — github.com/RishiiShah/Music
- Implemented a Bi-LSTM–based audio classification model using Librosa for MFCC, Chroma, and Mel-spectrogram features, achieving 98.73% accuracy.
- Optimized data preprocessing and training pipelines in Python, improving training efficiency by 22% and reducing memory usage by 15%.
- Deployed quantized model as a lightweight inference API for real-time genre tagging with low compute overhead.

Publications

Intelligent Traffic Surveillance: A Vision-Based System for Detecting Traffic Rule Violations — Sep. 2025
Developed a vision-based system using YOLOv12 and OCR to automate traffic violation detection, achieving 93.76% accuracy.

Bridging Financial Data Gaps with WGAN-GP Generating Synthetic Time Series for Robust Models — Aug. 2025
Synthesized realistic financial time series using WGAN-GP, improving predictive model accuracy by 0.9%.

JARVIS: Voice Assistant with Smart Home Automation — Jun. 2025
Designed and deployed a voice-controlled IoT automation system using LLMs and Raspberry Pi to simplify daily routines.